Integrating SAM With Feature Interaction for Remote Sensing Change Detection

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***Abstract*— Vision foundation models (VFMs) have rapidly gained application across various visual scenarios due to their robust universality and generalization capabilities. However, when directly applied to remote sensing images (RSIs), their performance often falls short owing to the unique inherent imaging characteristics. Moreover, these models typically suffer from inadequate feature extraction capabilities and unclear boundary detection because of the lack of specialized knowledge in the remote sensing (RS) field. To ameliorate these issues, we propose SFCD-Net, a novel network integrating SAM with feature interaction for RS change detection. To be specific, we first introduce a parameter-efficient fine-tuning (PEFT) method that allows the model to learn domain-specific knowledge, thereby enhancing its fine-grained feature extraction capability. Second, an innovative bitemporal feature interaction (BFI) module is designed to improve the model's sensitivity to changes. Finally, we use the boundary loss function (BLF) to enhance the model's ability to process boundary details, thereby improving its performance in recognizing boundaries and small targets. Through a series of ablation studies and comparative experiments, we demonstrate that the proposed SFCD-Net significantly improves model adaptability in RS tasks under limited computational resources, outperforming existing models.**

***Index Terms*— Change detection (CD), remote sensing (RS), vision foundation model (VFM).**

I. Introduction

REMOTE sensing change detection (RSCD) is at the forefront of research in the field of Earth observation, which analyzes multitemporal remote sensing images (RSIs) to identify and monitor surface changes [1]. These changes are typically represented as pixel-level binary classifications (change/no change) [2]. Comprehensive analysis of RSCD can facilitate decision-making in numerous intelligent applications [3], such as ecological change monitoring in environmental protection [4], [5], crop growth assessment in agriculture [6], land use change in urban planning [7], and emergency response in disaster management [8]. As a fundamental problem in RSI analysis, the change detection (CD) task has garnered significant attention from both academia and industry due to its immense practical application value.



Fig. 1. RS CD examples. Buildings a and b, which share the same semantics, differ in shape and appearance; buildings a and a#, and b and b#, differ in color and shadow due to factors such as illumination.

The high resolution of modern remote sensing (RS) technology allows for the capture of subtle details in surface features and changes, providing valuable information for land cover monitoring [9], [10]. However, this also introduces new challenges. RSI involves various types of ground objects with distinct characteristics [11], and imaging conditions such as illumination, shadows, and sensor types affect data quality [12]. These issues cause objects with the same semantics to exhibit different spectral characteristics at different locations and times [13]. For instance, in Fig. 1, buildings a and b, which share the same semantics, differ in shape and appearance; buildings a and a#, and b and b#, differ in color and shadow due to factors such as illumination. Therefore, to accurately identify changes in complex scenes, models need to comprehensively consider various factors, disregard nonsemantic changes caused by imaging conditions, and use appropriate feature representation methods to accurately capture the key characteristics of ground object changes [14].

In recent years, deep learning (DL) technology [15] has made significant advancements in the field of RSCD [16]. The CNN-based methods have achieved remarkable performance due to their robust feature extraction capabilities and stability [17]. A common approach involves using weight-sharing CNNs (Siamese CNNs [18]) to aggregate and segment semantic changes across multitemporal images [19]. The transformer-based methods leverage their global awareness capabilities, with encoders capturing long-range dependencies throughout the entire image [3], [12]. Despite these notable

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successes, their adaptability to complex and dynamic scenarios still falls short of practical application requirements. As model sizes increase, the limited availability of labeled data also significantly undermines the potential of these models [13].

Recently, vision foundation models (VFMs) have garnered significant attention due to their powerful generalization and adaptability [20]. These models (e.g., SAM [21] and CLIP [22]) are trained on extensive natural image datasets, thereby acquiring broad knowledge and representational capabilities. They exhibit robust generalization across different imaging conditions and visual objects, significantly advancing their applications in real-world scenarios [23], [24]. However, due to the inductive biases learned from natural images, VFMs face challenges when applied to RSI because of domain differences [25]. Compared with natural images, RSI typically features higher spectral resolution and greater geospatial complexity, which demand more precise segmentation capabilities from the models. Particularly, when images undergo significant changes, VFMs often struggle to effectively extract homogeneity and heterogeneity across multiple images [2]. This capability is crucial for CD tasks, as CD requires models to focus on regions with substantial changes [12].

To ameliorate the aforementioned issues, this article attempts to leverage the powerful and versatile semantic parsing capabilities of VFMs by integrating the general knowledge of SAM into the RSCD task, whose goal is to overcome domain knowledge differences and the challenges of detecting substantial changes in RSI. Specifically, we propose SFCD-Net, a novel network integrating SAM with feature interaction for RS change detection. First, we introduce a weight-decomposed low-rank adapter (DoRA) [26] into the SAM backbone network to learn domain-specific knowledge for RS, enhancing the model's fine-grained feature extraction capabilities. In addition, to better improve the model's sensitivity to changes, a bitemporal feature interaction (BFI) module is proposed to capture differences between images, thereby compensating for the model's ability to detect substantial changes. Finally, we design a boundary loss function (BLF) to improve the model's boundary perception, enhancing its performance in recognizing boundaries and small objects. Through a series of ablation studies and comparative experiments, we demonstrate that SFCD-Net can improve model adaptability in RSCD tasks under limited computational resources, significantly outperforming the existing techniques in terms of detection accuracy and operational efficiency. In summary, the main contribution can be summarized as follows.

1) We introduce a weight-DoRA into SAM backbone network to learn domain-specific knowledge for RS, which enhances the model's fine-grained feature extraction capabilities under limited computational resources.
2) A novel BFI module is designed to effectively capture feature differences between bitemporal RSI. This module ameliorates the model's shortcomings in homogeneity and heterogeneity, thereby highlighting substantial changes.
3) An improved BLF is incorporated to enhance the model's ability to capture boundary details, helping alleviate the issues of recognizing small changes and blurry boundaries and improving overall segmentation accuracy.

This article is organized as follows. We review the related work of DL-based RSCD methods and current VFMs in Section II. In Section III, we present our proposed SFCD-Net framework detailedly. Evaluation methods and extensive experimental results are shown in Section IV. Finally, we conclude this work in Section V.

II. Related Work

A. Deep-Learning-Based Methods

In the field of RSCD, deep-learning-based methods have demonstrated significant potential and advantages [20]. Initially, RSCD tasks based on convolutional neural networks (CNNs) were treated as segmentation problems [17], with network architectures such as UNet [27] becoming classical choices due to their ability to effectively integrate high-level and low-level features [28]. To further enhance performance, Siamese network structures were introduced into RSCD, using dual-branch networks to learn the differences between two images [18], [29]. This approach is particularly effective when handling paired images. In addition, to capture temporal dynamic changes in RSI sequences, the combination of recurrent neural networks (RNNs) and CNNs emerged as a new approach [30]. This combination leverages CNNs for spatial feature extraction while RNNs handle temporal sequence data, enabling the model to recognize and understand temporal variations in land features [30], [31]. Subsequently, transformer networks were introduced into RSCD due to their self-attention mechanism's advantage in handling long-range dependencies [12], [32]. Transformers can capture long-range dependencies across the entire image, providing the model with a global receptive field and opening new avenues for tasks requiring high-level semantic knowledge, such as RS change caption and RSCD [3], [14]. Despite the significant success of these methods, their adaptability to complex and dynamic scenarios still falls short of practical applications. As model scales increase, the limited availability of annotated data also significantly undermines the potential of these models.

B. Vision Foundation Models

Foundation models refer to models trained on extensive datasets [33]. In the field of computer vision, the concept of foundation models has gradually become key to addressing challenges in data requirements and generalization capabilities of deep learning [34]. These models are pretrained on large-scale datasets to acquire broad recognition capabilities, thereby adapting to various downstream tasks, especially data-intensive dense prediction tasks such as semantic segmentation [35] and CD [13]. CLIP [22], trained on a vast number of image-text pairs, has demonstrated performance in zero-shot image classification tasks comparable to fully supervised CNNs, which marks a significant achievement in the potential of foundation models for unsupervised [36] or few-shot learning [37]. In addition, researchers are exploring foundation models capable of handling vision tasks based on specific user prompts. For instance, the segment anything

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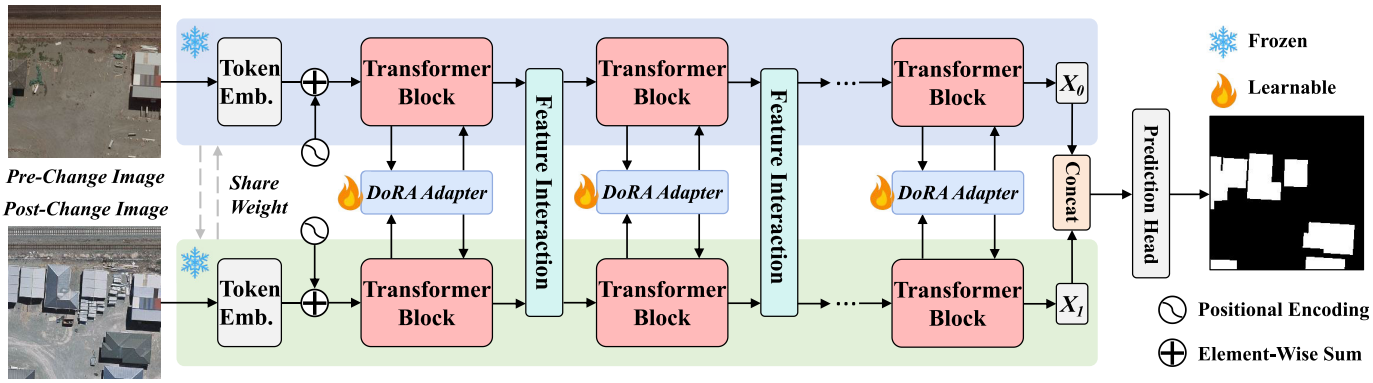


Fig. 2. Overall architecture of our proposed SFCD-Net. It mainly consists of three parts: the frozen image encoding transformer block, weight-DoRA, and the BFI module.

model (SAM) [21] is a segmentation model trained on millions of annotated images, capable of segmenting unseen images and objects during inference based on user prompts. SegGPT [38] attempts to learn visual concepts without explicit annotations by mimicking human learning processes. FastSAM [39] is proposed for real-time segmentation, significantly improving inference speed while maintaining comparable generalization performance to SAM. However, due to the inductive biases learned from natural images, VFMs face domain adaptation challenges when applied to RSI [2]. To overcome these limitations, this study introduces a fine-tuning method for VFMs to better learn and recognize latent semantic information in high-resolution RS imagery.

III. Methodology

In this section, we first formally introduce the problem definition of RSCD in Section III-A. Then, we present the overall architecture of our designed SFCD-Net in Section III-B. On this basis, we provide a detailed explanation of DoRA in Section III-C and the BFI module in Section III-D. Finally, in Section III-E, we introduce the loss function used during model training.

A. Problem Definition

Given two RSIs $T_0, T_1 \in \mathbb{R}^{H\times W\times C}$ with spatial resolution $H \times W$ and channel number C, the goal of the model is to predict a segmentation map S' with resolution $H \times W$ that approximates the ground truth (GT) S as closely as possible. Each pixel in S' belongs to one of the predefined classes $Y = \{y_0, y_1\}$, where y_0 represents the unchanged class and y_1 represents the changed class.

B. Model Architecture Overall

The overview of the SFCD-Net model is shown in Fig. 2. The input to the model is a set of bitemporal RSI. To accurately distinguish substantial changes between the two images, we leverage the general segmentation capability of the pretrained SAM model [21] to construct a CD network, SFCD-Net. To further enhance performance, we adopt a Siamese network [32] structure. Specifically, we introduce weight-DoRA in the image encoding module, allowing the model to maintain efficiency while learning domain-specific knowledge. For simplicity, both the branches of the network use the same pretrained architecture and are fine-tuned with the same adapter. The feature interaction module serves as a bridge between the two branches, embedded within the image encoding module, and is responsible for information exchange between image features at different temporal points. Through this interaction, the model can better capture and distinguish changes between different time points. Finally, we use a multilevel decoder to decode the features into the final CD results. To meet the requirements for fast and automatic processing, SFCD-Net does not require any prompts during inference. Therefore, we removed the prompt encoder, retaining only a decoder composed of a lightweight transformer and a segmentation head. Sections III-C–III-E will mainly introduce several key designs in SFCD-Net.

C. Weight-Decomposed Low-Rank Adapter

During the training process of RSI, parameter-efficient fine-tuning (PEFT) is used to update some parameters of the SAM model without updating the entire model. This not only saves computational resources but also reduces the difficulty of fine-tuning and storage, while ensuring good segmentation performance. Adapters are a type of small training module inserted between the layers of the pretrained model [13]. During training, the parameters of the pretrained model remain unchanged, and only the parameters of the adapter layers are updated. The low-rank approximation strategy, LoRA [40], is one of the mainstream adapter strategies. It uses the product of two small matrices to approximate the weight update

$$W' = W_0 + \Delta W = W_0 + AB \tag{1}$$

where $W_0 \in \mathbb{R}^{s\times t}$ is a layer parameter in the model, $\Delta W \in \mathbb{R}^{m\times n}$ is the update parameter matrix learned through back-propagation, and $A \in \mathbb{R}^{s\times r}$ and $B \in \mathbb{R}^{r\times t}$ are two low-rank matrices, with $r \ll s, t$.

However, Liu et al. [26] found that although LoRA can add or reduce width and direction updates proportionally, it lacks the ability to finely adjust parameters as a whole. Referring to this, we introduce the method to decompose the width and directional components, DoRA (Fig. 3), which first decomposes the pretrained weight matrix W_0 into a width

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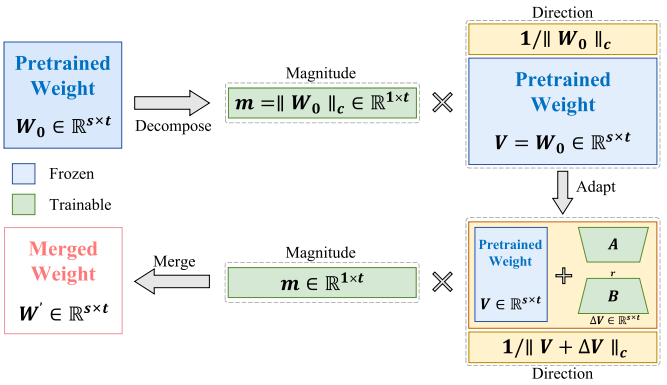


Fig. 3. Overview of the DoRA scheme, which decomposes the pretrained weight into magnitude and direction components for fine-tuning, especially with LoRA to efficiently update the direction component.

vector (m) and a directional matrix (V) with $\| \cdot \|_c$ being the vectorwise norm of a matrix across each column

$$W_0 = m \frac{V}{\|V\|_c}. \tag{2}$$

Then, using the initial weight matrix W_0 as the directional matrix ($V = W_0$), the width vector m is trained independently through LoRA

$$W' = m \frac{V + \Delta V}{\|V + \Delta V\|_c} = m \frac{W_0 + AB}{\|W_0 + AB\|_c} \tag{3}$$

where only m, A, and B are learnable parameters. In practice, we introduce DoRA in each layer of the multihead attention in the encoder. The weight matrix is decomposed into two parts: magnitude and direction. Subsequently, a bypass composed of downprojection and upprojection is incorporated. The downprojection compresses the given embedding to a lower dimension, while the upprojection expands the compressed embedding back to its original dimension. Based on this, when the input feature F is fed in, the processing strategy of the self-attention mechanism becomes

$$Q = W_q' F, \quad K = W_k' F, \quad V = W_v' F \tag{4}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V. \tag{5}$$

D. Bitemporal Feature Interaction Module

Current VFMs are not usually designed for CD [41]. They are often adept at processing single images but lack the ability to extract homogeneous and heterogeneous features when dealing with multiple images [2]. This poses a challenge for RSCD task, as a robust CD model must be capable of identifying substantial changes. To address this issue, we designed a BFI module, as shown in Fig. 4. The BFI projects bitemporal features into a feature semantic space for interaction, enabling the foundation model to detect "effective changes."

Assume that at a certain stage of the model, the encoder receives the encoded image features F_0 and F_1 at corresponding times T_0 and T_1. First, the absolute value difference matrix D of the elements at the same position is calculated

$$D = \text{Conv}(|F_0 \ominus F_1|) \tag{6}$$

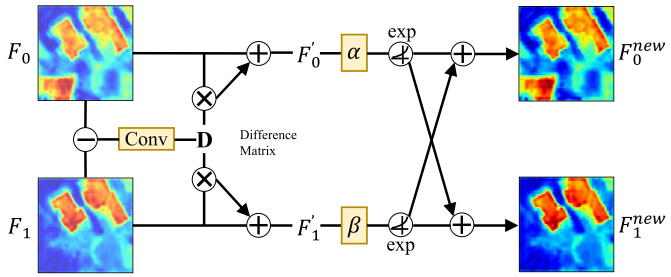


Fig. 4. Overview of the BFI module. The BFI interacts with bitemporal features to enhance the model's ability to detect substantial changes.

where $|\cdot|$ denotes the absolute value operation, $\ominus$ denotes the elementwise subtraction operation, D represents the dual feature difference matrix, and Conv($\cdot$) represents the convolution operation with batch normalization and ReLU activation function. Considering that D can be regarded as the attention weight, the module passes the elementwise multiplication operation to the temporal features to highlight the changing regions of the features. Then, the enhanced features are added to the original features ($\oplus$ denotes the elementwise addition operation) to further optimize the feature representation to obtain F_0' and F_1'

$$F_0' = D \oplus F_0, \quad F_1' = D \oplus F_1. \tag{7}$$

In the second interaction, F_0' and F_1' are passed through two different 1D convolution modules α and β to project into the latent state, and then converted back into the original feature dimension in the exp format and added to the original features. Therefore, the final output of the BFI module is two updated subfeatures F_0^{new} and F_1^{new}

$$F_0^{\text{new}} = F_0 \oplus \exp\left(\alpha\left(F_0'\right)\right) \tag{8}$$

$$F_1^{\text{new}} = F_1 \oplus \exp\left(\beta\left(F_1'\right)\right). \tag{9}$$

The backbone of the SAM [21] consists of multiple transformer encoder blocks, the number of which can be chosen based on requirements, including the base version (12 blocks), large version (24 blocks), and huge version (32 blocks). To improve computational efficiency, most of the transformer encoder blocks use a local attention mechanism, where each position only interacts with its nearby positions rather than the entire sequence. To capture global information, SAM uses a global attention mechanism in a few selected layers. SFCD-Net incorporates the BFI module after these global attention layers. This approach not only reduces the number of parameters and computational load but also generates multilevel temporal difference features to capture rich contextual information from temporal images.

E. Boundary Loss Function

In image segmentation tasks [35], the commonly used loss functions include cross-entropy (CE) loss and Dice loss, both of which are region-based metrics. CE loss is used for pixel-level classification

$$\mathcal{L}_{CE} = -\sum_{i=1}^{N} y_i \log(\omega_i) \tag{10}$$

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where N is the total number of pixels, y_i is the true label of the ith pixel, and ω_i is the predicted probability of the ith pixel. Dice loss is used to measure the overlap between the predicted results and the true labels

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \omega_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \omega_i}. \tag{11}$$

However, since the CE loss function treats all the pixels equally, it focuses on overall classification accuracy; the Dice loss ignores spatial information and fails to penalize missegmentation far from the true labels more heavily. As a result, segmentation boundaries are often blurred, and smaller targets are not recognized. To address these issues, referring to [42], we introduce an improved BLF, which uses distance measurement in the contour space rather than the region space. Specifically, let Ω represent the spatial domain of the image, $\lambda : \Omega \to \{0, 1\}$ be the binary map of the true segmentation of the image, and $h : \Omega \to \{0, 1\}$ be the corresponding predicted segmentation binary map. $G = \{\omega \in \Omega | \lambda(\omega) = 1\}$ is the GT change region, and $H = \{\omega \in \Omega | h(\omega) = 1\}$ is the predicted change region. The circular region represents G, and the irregular region represents H. The goal is to construct a BLF $\text{Dist}(\partial G, \partial H)$, which measures the distance metric in the boundary space of the region in Ω

$$\text{Dist}(\partial G, \partial H) = \int_{\partial G} \| y_{\partial H}(p) - p \|^2 dp \tag{12}$$

where $p \in \partial G$ is a point on the boundary ∂G, and $y_{\partial H}(p)$ is the corresponding point on boundary ∂H. Through integral transformation and approximation, we get

$$\text{Dist}(\partial G, \partial H) \approx 2 \int_{\Delta H} D_G(q) dq \tag{13}$$

where $D_G(q)$ is a distance map relative to the boundary ∂G, representing the distance between point q and the nearest point on ∂G. Then, further derivation gives

$$\frac{1}{2}\text{Dist}(\partial G, \partial H) = \int_H \phi_G(q) dq - \int_\Omega \phi_G(q) \lambda(q) dq \tag{14}$$

where ϕ_G is the level set representation of the boundary ∂G. After simplification, the final BLF is

$$\mathcal{L}_{\text{BLF}} = \int_\Omega \phi_G(q) h(q)\, dq. \tag{15}$$

SFCD-Net combines CE loss and boundary loss, ensuring overall pixel classification accuracy while capturing boundary information. Hence, the total loss is

$$\mathcal{L}_{\text{total}} = \eta \mathcal{L}_{CE} + (1 - \eta) \mathcal{L}_{BLF} \tag{16}$$

where η is a hyperparameter used to balance the weights of two loss functions.

IV. Experiments

In this section, we first describe the dataset, evaluation metrics, and experimental setup details in Section IV-A and Section IV-B. Furthermore, Section IV-C introduces the SOTA approaches and provides comparisons of retrieval performance. Finally, we perform sufficient ablation experiments to further analyze in Section IV-D.

A. Dataset and Evaluation Metrics

1) Datasets: We evaluated our proposed SFCD-Net framework on two widely used RSCD datasets: LEVIR-CD [43] and WHU-CD [44]. To maintain a fair and consistent comparison, we used the same dataset partitions, described as follows.

a) LEVIR-CD [43]: A publicly available large-scale dataset for RSCD, consisting of 637 pairs of ultrahigh-resolution (0.5 m/pixel) image patches, each with a size of 1024 × 1024 pixels. This dataset covers 20 different areas across multiple cities in the United States and includes significant seasonal and illumination variations. Within the dataset, there are 31 066 643 change pixels, accounting for 4.65% of the total, and 636 876 269 unchanged pixels, making up 95.35% of the total. Hence, this is an extremely imbalanced binary classification dataset (with change pixels labeled as 1 and unchanged pixels as 0). We adopt the default data split (training set: 445 pairs of images; validation set: 64 pairs of images; test set: 128 pairs of images), which is published on the authors' website.[1]

b) WHU-CD [44]: A publicly available RSCD dataset, containing a pair of ultrahigh-resolution (0.2 m/pixel) image patches, sized 32,507 × 15,345, which are cropped into 512 × 512 patches. This dataset covers an area of New Zealand and includes 21 442 501 change pixels, accounting for 4.26% of the total, and 481 873 979 unchanged pixels, making up 95.74% of the total with change pixels labeled as 1 and unchanged pixels as 0. We adopt the default data split (training set: 1260 pairs of images; test set: 690 pairs of images), which is published on the authors' website.[2] To ensure a fair comparison, we directly crop the default image pairs into nonoverlapping 256 × 256 patches and randomly select 10% of the training data to form a validation dataset. Consequently, we obtain a dataset comprising 4536/504/2760 pairs of image patches for training/validation/testing, respectively.

2) Evaluation Metrics: To quantitatively validate the effectiveness of our proposed model, we use the following metrics for evaluation: $F1$ score (F1), precision (Pre.), recall (Rec.), overall accuracy (OA), and intersection over union (IoU). These metrics are used by comparing the GT with the predicted maps and are defined as follows:

$$F1 = 2 \cdot \frac{\text{Pre.} \cdot \text{Rec.}}{\text{Pre.} + \text{Rec.}} \tag{17}$$

$$\text{Pre.} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{Rec.} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{20}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{21}$$

where TP, TN, FP, and FN are the numbers of true positives, true negatives, false positives, and false negatives, respectively.

[1] http://chenhao.in/LEVIR

[2] http://gpcv.whu.edu.cn/data/building_dataset.html

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B. Implementation Details

SFCD-Net uses the vision pretraining model of SAM based on ViT-Large for foundation knowledge learning. During the stage of weight decomposition low-rank fine-tuning, we set the rank to 16 to maintain model consistency. To ensure the efficiency of the image decoder and prediction head, we limit the image upsampling to (1/4) of the original image when training on the LEVIR-CD dataset. SFCD-Net is trained using binary CE loss and BLFs, with η set to 0.9. We set different input sizes based on different datasets and use data augmentation techniques such as rotation, flipping, random cropping, and photometric distortion to increase the training sample size. During the training phase, the backbone of SAM remains frozen while other parts are trainable. We use the AdamW optimizer with an initial learning rate of 1e-4 and use warmup strategy to adjust it automatically during training. When training on LEVIR-CD, the batch size is set to 4, and for WHU-CD, it is set to 8, ensuring training on a single NVIDIA RTX4090. We set the training epoch of SFCD-Net to 300 and select the best-performing model on the validation set for testing and evaluation.

C. Comparisons to State-of-the-Art Methods

We compared the proposed SFCD-Net with a series of advanced RSCD methods, including CNN-based methods (HANet [9], CBSASNet [18], GCD-DDPM [16], WNet [29]), transformer-based methods (BIT [12], ChangeFormer [32], VcT [14], M-Swin [3]), and foundation model-based methods (BAN [45], ChangeCLIP [41], SAM-CD [46]). The experimental results of other methods presented in this article are based on the results reported in their original studies. A more detailed analysis of these methods is as follows.

BIT[3]: A transformer-based approach that improves the contextual information of CNN features using a transformer encoder–decoder network with semantic tokens, and then uses feature differencing to produce the change map.

ChangeFormer[4]: A transformer-based Siamese network architecture that efficiently captures multiscale, long-range detailed information through the use of a hierarchical transformer encoder and a lightweight MLP decoder.

HANet[5]: A progressive foreground balanced sampling strategy, which aids the model in accurately learning foreground image features by initially focusing on the foreground and then gradually incorporating the background.

VcT[6]: Combining a shared backbone and GNN to extract and model structured information, and leveraging cross-attention and anchor-primary attention module to refine reliable tokens for accurate prediction.

WNet[7]: A dual Siamese branching encoder that integrates both CNN and transformer components to capture local and global information and incorporates deformable convolution and attention mechanisms to focus on critical regions.

CBSASNet: A Siamese network that uses channel bias split attention to recover change region information and uses cross-temporal fusion module to highlight it through the optimized single-temporal image features.

GCD-DDPM[8]: A generative CD model, which leverages DDPM to generate CD maps directly and uses a difference conditional encoder (DCE) to guide this process using multilevel difference features.

M-Swin [3]: This method uses a novel Siamese encoder to capture change information and fuse multiscale features and a bitemporal image feature fusion module is introduced to enhance features and improve the saliency of change regions.

BAN[9]: A bitemporal adapter network using CLIP, a bitemporal adapter branch (Bi-TAB) and bridging modules. CLIP extracts general features, which are injected into Bi-TAB to extract task-specific features, compatible with any CD model or custom blocks.

ChangeCLIP[10]: A framework leveraging semantic information from image–text pairs. It reconstructs CLIP to extract bitemporal features and uses a differential features compensation module to capture detailed semantic changes.

SAM-CD: A framework using SAM's semantic knowledge with multiscale feature extraction and masked attention to fuse local and global features for accurate CD.

Table I presents the performance of different methods on the LEVIR-CD and WHU-CD datasets. After comparing the performance of these methods, we find that SFCD-Net excels in multiple key metrics, indicating its high accuracy and stability in the RSCD task. Specifically, SFCD-Net achieved SOTA in almost all the metrics. Notably, in terms of $F1$ and IoU metrics, SFCD-Net significantly outperformed the current advanced methods ChangeCLIP (92.01/85.20) and SAM-CD (92.24/85.99) on the LEVIR-CD dataset (92.59/86.19). Similarly, on the WHU-CD dataset, SFCD-Net (95.23/90.89) also significantly surpassed ChangeCLIP (94.82/90.15) and SAM-CD (95.13/90.71). Moreover, we observe that the overall accuracy levels of the models are close, which we believe is because overall accuracy is easily influenced by class balance and sample size. In the LEVIR-CD and WHU-CD datasets, there is a significant difference in the number of samples across different classes, with the changed part being much fewer than the unchanged part. Consequently, models perform better on the more numerous classes, resulting in higher overall accuracy. Overall, models based on VFM outperform those based on CNN and transformer, which we attribute to the strong feature extraction capability of the base model. The results demonstrate that our proposed SFCD-Net performs the best, which validates the rationality of each module we designed.

In the experimental results, we observed that the number of parameters and the number of floating-point operations (Flops) significantly impact model performance, but they are not the sole determining factors. Despite BIT having the smallest number of parameters (3.50 M) and Flops (10.63 G),

[3]https://github.com/justchenhao/BIT_CD
[4]https://github.com/wgcban/ChangeFormer
[5]https://github.com/ChengxiHAN/HANet-CD
[6]https://github.com/Event-AHU/VcT_Remote_Sensing_Change_Detection
[7]https://github.com/TangXu-Group/Remote-Sensing-Image-Change-Detection/tree/main/WNet
[8]https://github.com/udrs/GCD
[9]https://github.com/likyoo/BAN
[10]https://github.com/dyzy41/ChangeCLIP

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TABLE I

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE LEVIR-CD AND WHU-CD DATASETS

Methods	Reference	Type	Params (M)	Flops (G)	LEVIR-CD					WHU-CD				
					Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
BIT [12]	*TGRS'21*	Transformer	3.50	10.63	89.24	89.37	89.31	80.68	98.92	86.64	81.48	83.98	72.39	98.75
ChangeFormer [32]	*IGARSS'22*	Transformer	13.94	26.42	92.05	88.80	90.40	82.48	99.04	91.83	88.02	89.88	81.63	99.12
HANet [9]	*J-STARS'23*	CNN	3.03	14.07	91.21	89.36	90.28	82.27	99.02	88.30	88.01	90.20	78.82	99.16
VcT [14]	*TGRS'23*	Transformer	3.57	10.64	92.57	87.65	90.04	81.89	99.01	89.39	89.77	89.58	81.12	99.18
WNet [29]	*TGRS'23*	CNN	43.07	19.20	91.16	90.18	90.67	82.93	99.06	92.37	90.15	91.25	83.91	99.31
CBSASNet [18]	*TGRS'24*	CNN	5.76	31.64	92.47	90.33	91.39	84.14	99.13	93.93	91.15	92.52	86.08	99.31
GCD-DDPM [16]	*TGRS'24*	CNN	130.83	269.52	90.68	**91.24**	90.96	83.65	99.14	92.79	92.29	92.54	86.52	99.33
M-Swin [3]	*TGRS'24*	Transformer	-	-	92.16	89.77	90.95	83.30	99.09	93.89	88.65	91.19	83.81	99.32
BAN [45]	*TGRS'24*	VFM	3.80	330.32	**92.83**	90.89	91.85	84.93	99.12	94.55	**93.73**	93.99	88.68	99.48
ChangeCLIP [41]	*ISPRS'24*	VFM	-	-	**93.40**	90.67	**92.01**	**85.20**	**99.20**	**95.63**	**94.02**	**94.82**	**90.15**	**99.52**
SAM-CD [46]	*ICASSP'24*	VFM	-	-	**93.58**	**90.93**	**92.24**	**85.59**	**99.22**	**96.67**	93.64	**95.13**	**90.71**	**99.62**
SFCD-Net	Ours	VFM	14.56	273.43	92.79	**92.40**	**92.59**	**86.19**	**99.21**	**95.50**	**94.95**	**95.23**	**90.89**	**99.63**

All values are in %. Higher F1 and IoU values indicate better CD performance. **Red**, **blue**, and **bold** respectively denote the best, 2nd-best, and 3rd-best.

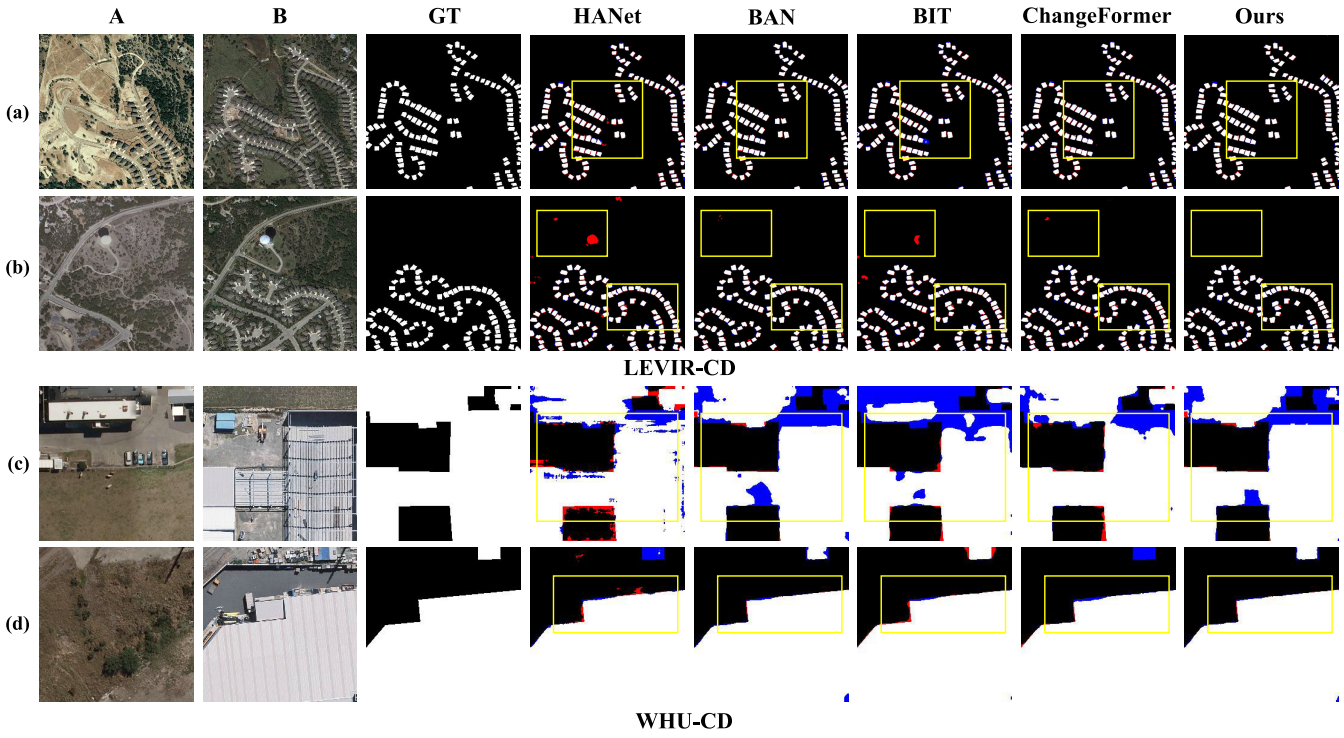


Fig. 5. Visualization comparison of different methods on the (a) and (b) LEVIR-CD. (c) and (d) WHU-CD test sets. For convenience, several colors are used to facilitate a clearer visualization of results, i.e., TP (white), FP (red), TN (black), and FN (blue).

its performance is not outstanding. Conversely, GCD-DDPM and M-Swin have the highest Flops (269.52 and 330.32 G, respectively), yet their performance is also suboptimal. The best-performing model, SFCD-Net, achieved the highest $F1$ and IoU scores on the WHU-CD dataset. Although its parameters and Flops (14.56 M and 273.43 G) are not the smallest (as it is based on VFM), it demonstrates a clear advantage in performance, showcasing its excellent ability to balance computational complexity and performance.

Fig. 5 presents the visualization comparison results of different methods on the LEVIR-CD and WHU-CD datasets. We selected some challenging samples (a)–(d) with complex building features from the test set for illustrative purposes. This figure intuitively shows comparison of the performance of each model. To further enhance the intuitiveness of the visualization, we used different colors and red boxes to show the key and detailed results of the models, where blue and red indicate missed detections and false detections, respectively. Specifically, SFCD-Net performs better in avoiding missed and false detections of complex building features [Fig. 5 (a) and (b)]. BiT and ChangeFormer, in particular, have obvious missed detection issues. Compared with the CNN-based methods HANet, SFCD-Net has significantly fewer blue and red pixels. In addition, SFCD-Net excels in boundary handling and small object detection. By observing the red areas next to the buildings in Fig. 5(c) and (d), it can be seen that SFCD-Net is more sensitive to detailed changes compared with other methods such as BAN.

D. Ablation Study

This section conducts ablation experiments by incrementally adding different modules to the SAM model to observe and compare the model's performance, thereby validating the positive impact of the proposed strategies on the model's performance. Specifically, to further verify the effectiveness of each module, we individually removed the DoRA adapter,

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TABLE II
PERFORMANCE OF SFCD-Net AND ITS VARIANTS ON THE LEVIR-CD AND WHU-CD DATASETS

Method	DoRA	BFI	BLF	Params (M)	LEVIR-CD					WHU-CD				
					Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
Variant1				7.03	85.02	82.12	83.54	71.74	98.35	90.50	83.70	86.97	76.94	99.02
Variant2	✓			14.10	92.40	90.35	91.37	84.11	99.13	94.78	92.57	93.67	88.09	99.51
Variant3	✓	✓		14.56	**93.12**	91.74	92.42	85.91	**99.23**	**95.63**	93.80	94.71	89.95	99.59
SFCD-Net	✓	✓	✓	14.56	92.79	**92.40**	**92.59**	**86.19**	99.21	95.50	**94.95**	**95.23**	**90.89**	**99.63**

All values are in %. Higher F1 and IoU values indicate better CD performance. **Red** denote the best.

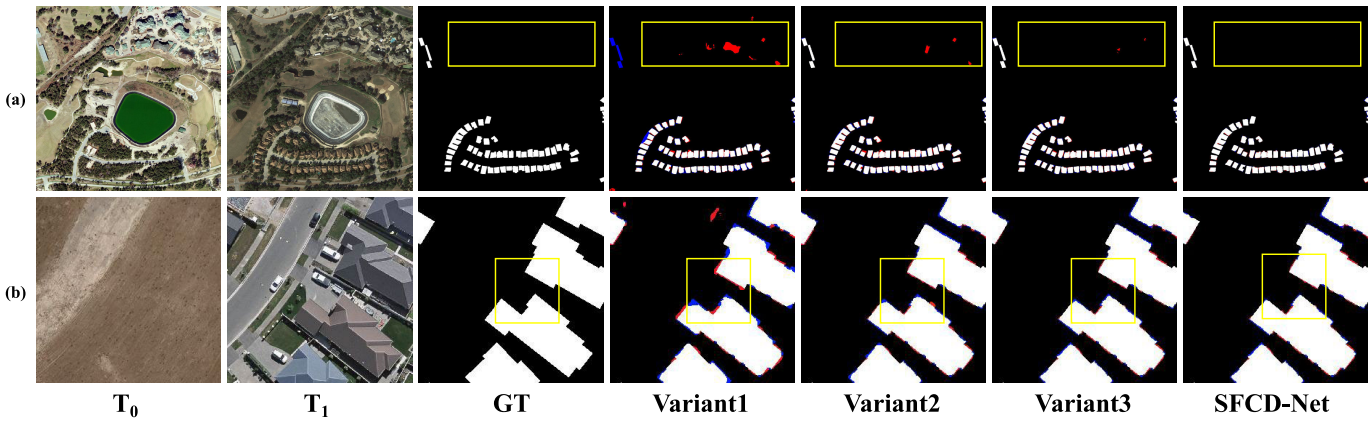


Fig. 6. Visualization comparison of different variants on (a) LEVIR-CD and (b) WHU-CD test sets. For convenience, several colors are used to facilitate a clearer visualization of results, i.e., TP (white), FP (red), TN (black), and FN (blue).

the BFI module, and the BLF, and compared the performance with the complete SFCD-Net model. In addition, we explored different parameter settings within the modules.

1) Different Modules: The baseline model is the pretrained SAM model, where only the multilevel decoder head is trainable, while all other parameters are frozen, and the loss function is the CE function. Table II shows the impact on detection rate and performance after adding the aforementioned improvement strategies to the baseline model. The results indicate that the overall performance of the model progressively improves with the introduction of different strategies.

Specifically, by comparing Variant 1 and Variant 2, it can be observed that the introduced fine-tuning strategy significantly improves the performance of SAM in RSCD. Notably, the trainable parameters in Variant 2 account for only 0.52% of the backbone. By adding a small number of parameters for fine-tuning to SAM, SFCD-Net bridges the gap between natural image segmentation and RSCD with minimal system overhead. This finding indicates that a fine-tuned strategy can effectively enhance the model's performance in specific tasks while maintaining system efficiency and lightness. Variant 3 outperforms Variant 2 in all the metrics, demonstrating the effectiveness of the BFI module. The trainable parameters in Variant 3 account for only 3.17% of the backbone. Although this increases the parameter count to some extent, it remains a deployable solution compared with full-parameter tuning. When pursuing CD quality, this additional overhead is worthwhile. SFCD-Net shares the same structure as Variant 3 but introduces a BLF during training. It performs better in recall, $F1$, and IoU metrics, although slightly lower in precision. A lower precision often indicates an increase in false positives, while a higher recall suggests a reduction in false negatives. The $F1$-score, being the harmonic mean of the two, not showing significant change is reasonable. The improvement in IoU indicates that the model is more accurate in spatial localization; even with more false positives, these false positives are closer to the actual targets. These results suggest that SFCD-Net significantly improves in detecting more targets and achieving more precise localization, greatly addressing issues of boundary blurring and small target recognition difficulties.

Fig. 6 presents the visual results of different variants on the LEVIR-CD and WHU-CD datasets, providing an intuitive comparison of their performance. Specifically, Variant 1, which only uses the SAM pretrained model, clearly exhibits instances of missed detections and false positives and is insensitive to the changes in the region. This could be due to the lack of RS domain-specific knowledge in the pretrained models for natural images. Variant 2, after introducing the DoRA adapter, shows a significant reduction in red and blue pixels, resulting in a notable improvement in CD, although some false positives remain. With the introduction of all the modules, the fully equipped SFCD-Net outperforms other variants in handling boundary and small target detection issues. This not only demonstrates that our method can effectively learn domain-specific knowledge but also highlights its superior performance in detailed CD, proving the effectiveness of the BFI module and BLF.

2) DoRA Rank: The key parameter in the DoRA layer is the rank, as different ranks affect RSCD performance. Tables III and IV show the impact of different ranks on the LEVIR-CD and WHU-CD datasets, respectively. On the

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TABLE III
INFLUENCE OF DoRA RANK IN SFCD-Net ON THE LEVIR-CD DATASET

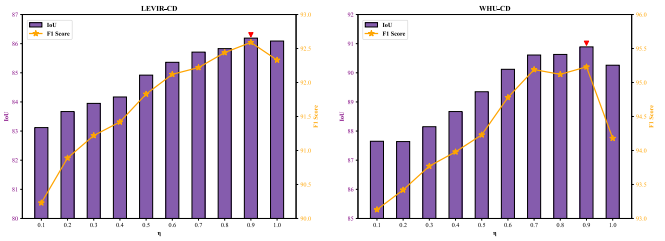
Rank	LEVIR-CD				
	Pre.	Rec.	F1	IoU	OA
1	92.33	90.51	91.41	84.18	98.80
2	92.45	91.24	91.84	84.91	99.17
4	93.18	90.94	92.05	85.26	99.20
8	92.94	91.75	92.34	85.77	99.22
16	92.79	**92.40**	**92.59**	**86.19**	99.21
32	92.69	92.36	92.53	86.09	**99.24**
64	**93.39**	91.50	92.43	85.93	**99.24**
128	93.07	91.42	92.24	85.59	99.22

TABLE IV
INFLUENCE OF DoRA RANK IN SFCD-Net ON THE WHU-CD DATASET

Rank	WHU-CD				
	Pre.	Rec.	F1	IoU	OA
1	94.16	92.72	93.44	87.68	99.49
2	95.18	92.27	93.70	88.15	99.51
4	**96.23**	94.00	95.11	90.67	99.62
8	95.91	94.24	95.07	90.60	99.62
16	95.50	94.95	95.23	90.89	99.63
32	95.77	94.81	95.28	90.99	99.63
64	95.56	95.18	95.37	**91**.15	**99.64**
128	95.61	**95.26**	**95.43**	**91.26**	**99.64**

LEVIR-CD dataset, we find that the detection performance of SFCD-Net continues to improve within a certain range of DoRA ranks. Smaller ranks are unable to adapt to the RSCD task of SAM. After Rank = 16, the detection performance of SFCD-Net fluctuates. Larger ranks increase the number of training parameters, but do not significantly improve performance compared with Rank = 16. We believe this is because the adapter layer requires a basic number of parameters to adapt to the characteristics of the RSI dataset, but too many trainable parameters may weaken the model's original knowledge, thereby increasing training difficulty. On the WHU-CD dataset, we find that the detection performance of SFCD-Net gradually improves with the increase in DoRA rank, but the improvement slows down after Rank = 16. We believe this is because the WHU-CD dataset has more samples, and the model continues to learn. Considering the overall model performance, to balance model size and detection performance, we ultimately set the rank of SFCD-Net to 16.

3) Balance Parameter η: In our SFCD-Net, there is a hyperparameter η that balances the contributions of the two terms in (16). To investigate its impact on the model, we adjust its value from 0.1 to 1.0 in increments of 0.1. Then, we calculate the $F1$ and IoU values on the LEVIR-CD and WHU-CD datasets and display them in Fig. 7. By observing these figures, we find the following points. First, when η increases from 0.1 to 0.9, the model performance gradually improves, indicating that the boundary loss L_{BLF} is relatively more important. However, when $\eta = 1.0$, the performance of SFCD-Net significantly decreases, indicating that the CE loss



(a) (b)

Fig. 7. $F1$ score and IoU of SFCD-Net with different η on (a) LEVIR-CD and (b) WHU-CD.

is indispensable. In addition, we find that the performance variability of SFCD-Net on the WHU-CD dataset is more pronounced than on the LEVIR-CD dataset. This is mainly because the distribution of change instances differs, but these variations are within a reasonable range.

V. CONCLUSION

In this article, we propose an innovative network, SFCD-Net, which integrates SAM with feature interaction for RSCD. By introducing PEFT methods, we can inject domain-specific knowledge into the model, enhancing its ability to extract fine-grained features. In addition, the novel bitemporal feature interaction (BFI) module significantly improves the model's sensitivity to changes. We also use a BLF to enhance the model's ability to handle boundary details, resulting in notable improvements in identifying boundaries and small targets. Through a series of ablation studies and comparative experiments, we demonstrate that the proposed SFCD-Net significantly improves the model's adaptability to RS tasks under limited computational resources, outperforming the existing models.

However, this article represents merely an initial attempt to transfer VFM to a specific task, and there remain many shortcomings that need to be addressed. Although the introduction of BLF has provided some improvement in model accuracy, its effect is limited. To enhance the recognition ability of small objects and address issues such as unclear boundary discrimination, it is necessary to strengthen feature extraction and semantic understanding capabilities. Future research should focus on better understanding instance edge information and the semantics within small targets.

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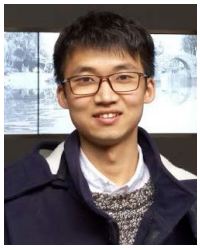
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